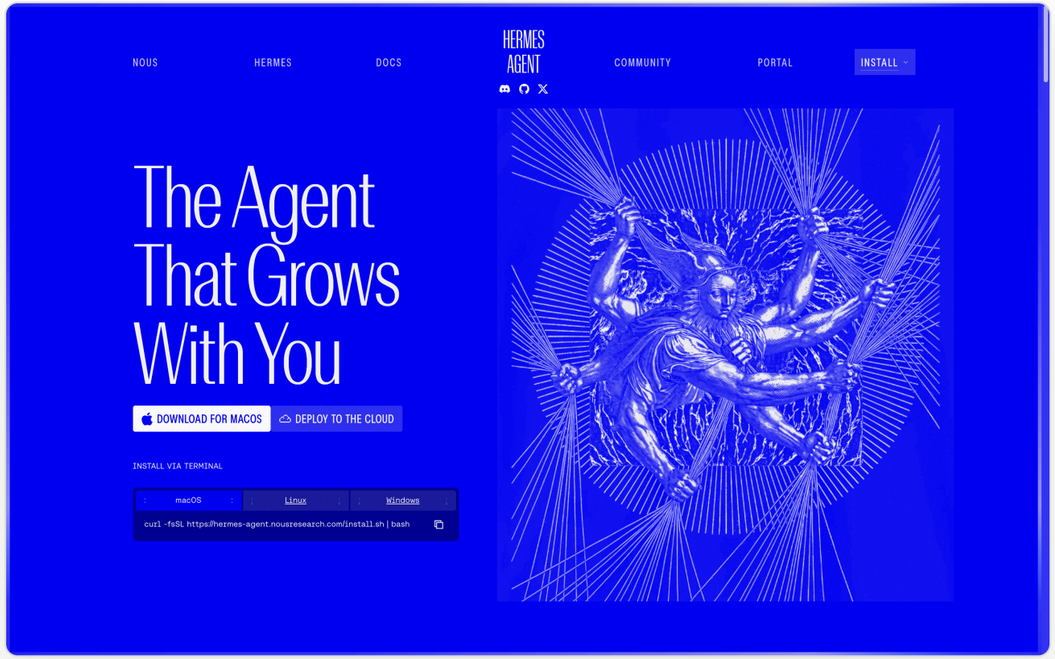
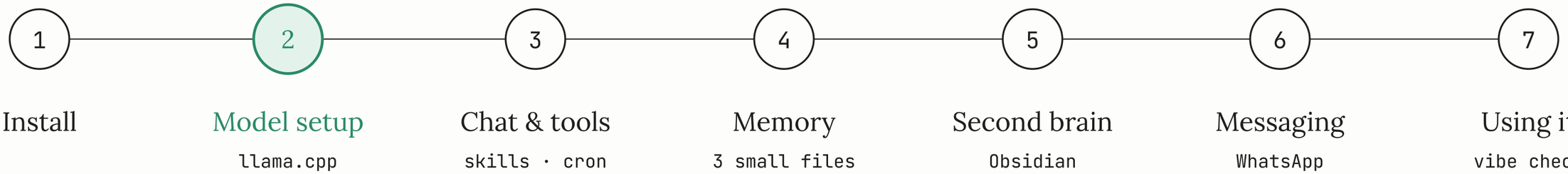
Build your first local agent with Hermes

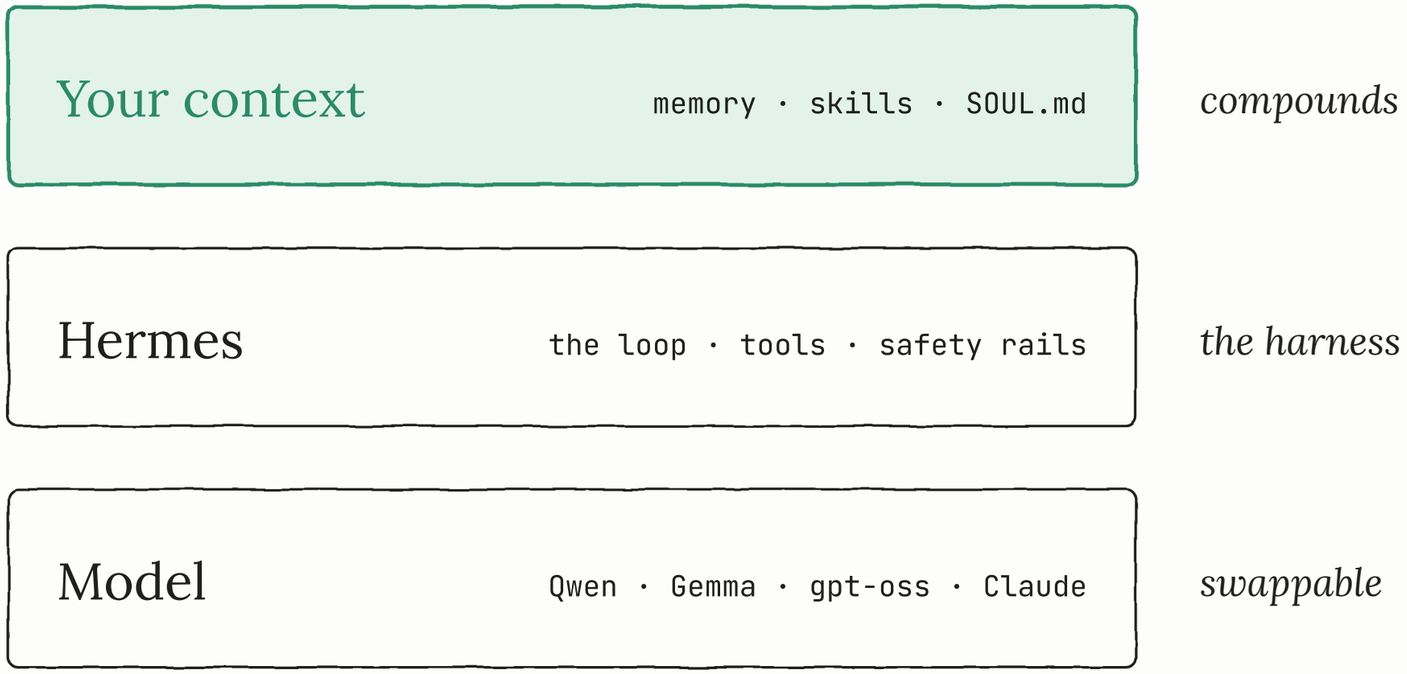
A private agent that runs on your machine, remembers you, and works while you sleep.



Today


1
2
3
4
5
6
7
Install
Model setup
llama.cpp
Chat & tools
skills · cron
Memory
3 small files
Second brain
Obsidian
Messaging
WhatsApp
Using it
vibe check

Three layers



Swap the model. Keep the context.

Install

```
curl -fsSL https://hermes-agent.nousresearch.com/install.sh | bash
```

```
hermes --version
Hermes Agent v0.22.1 (2026.9.24)
```

Windows: the PowerShell one-liner
Mac: the Desktop app works too

Without Hermes Desktop:

For a command-line only install without Hermes Desktop, run:

Linux / macOS / WSL2

```
curl -fsSL https://hermes-agent.nousresearch.com/install.sh | bash
```

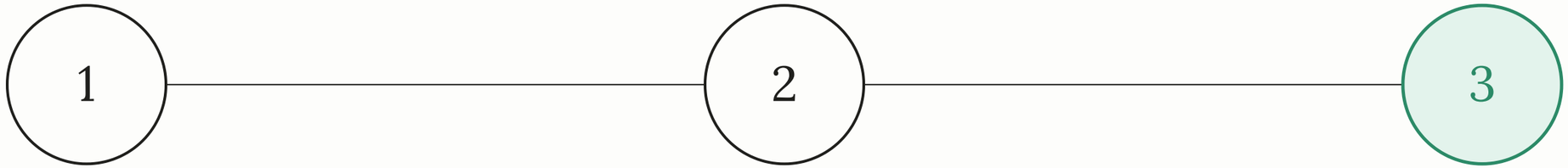
Windows (native)

Run in powershell:

```
iex (irm https://hermes-agent.nousresearch.com/install.ps1)
```

the part everyone gets stuck on

Model setup is three decisions



Engine

`llama.cpp`

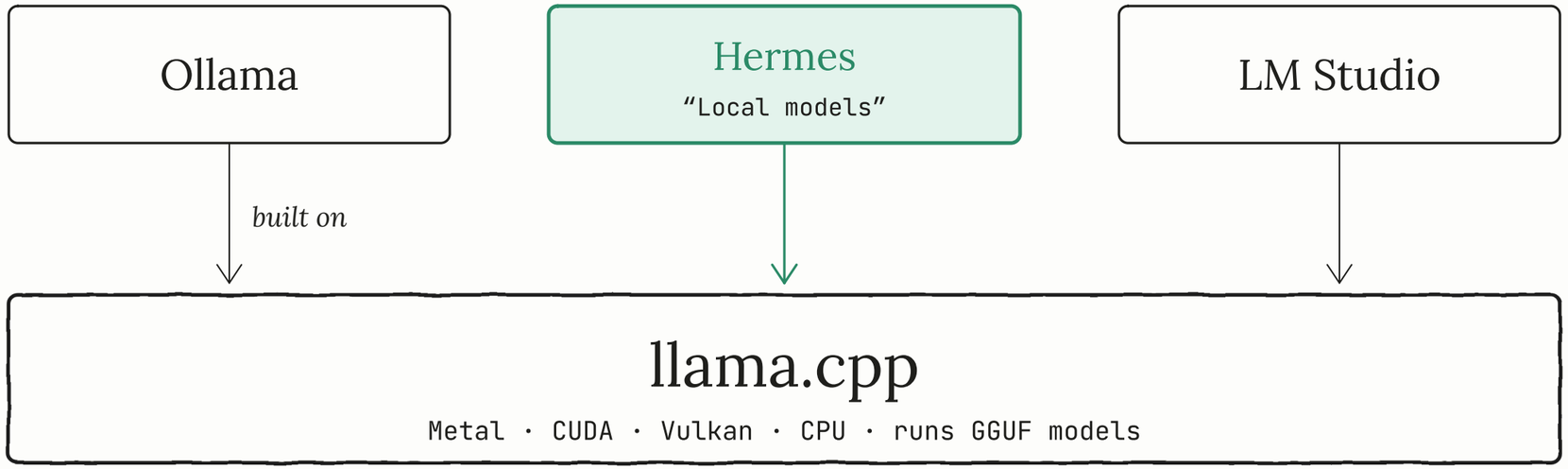
A model that fits

`your RAM / VRAM`

Plug it into Hermes

`hermes model`

1 · The engine: llama.cpp



also good

MLX / oMLX

Apple only · MLX models

vLLM

NVIDIA servers, many users

`OpenAI-compatible`

Hermes talks to `localhost:8080/v1`

`--jinja`

tool calls through the model's chat template

`-c 65536`

you control the context window

llama.cpp server README · Ollama & LM Studio docs · Hermes local-models docs

2 · Will it fit? Do the math once.

weights + KV cache + ~1 GB ≤ GPU memory

`weights`

params × bits ÷ 8

Q4_K_M ≈ 0.6 GB per billion params

`kv cache`

grows with the context

Hermes needs ≥ 64K tokens

`gpu memory`

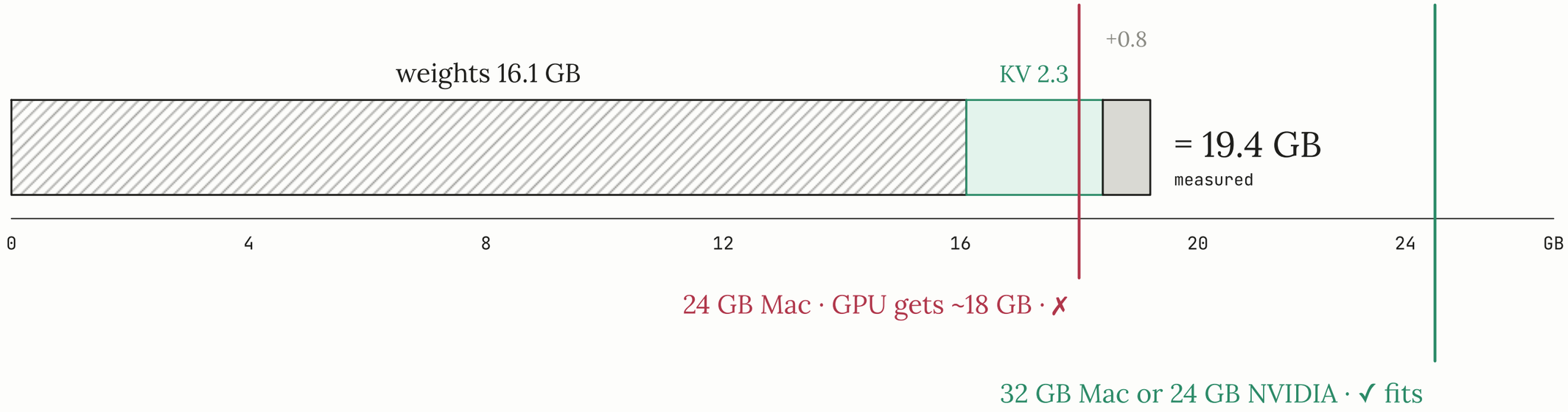
Mac ≈ 75% of RAM

NVIDIA: your VRAM

Q4_K_M = 4.89 bits/weight (llama.cpp quantize README) · Mac GPU limit: Metal recommendedMaxWorkingSetSize

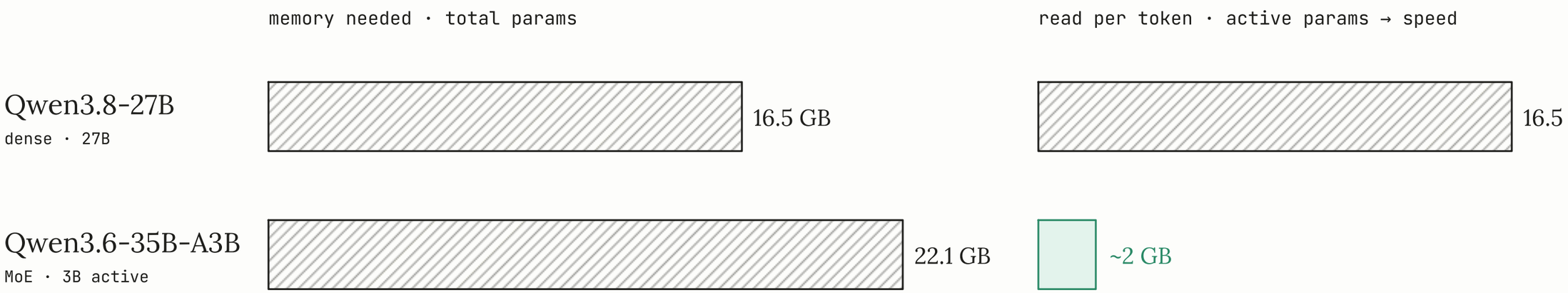
example · Qwen3.8-27B · UD-Q4_K_M · 64K context · q8_0 KV cache

Does it fit on my laptop?



measured with llama-fit-params on llama.cpp b11429 · unsloth/Qwen3.8-27B-GGUF

Dense vs MoE



MoE: more memory, much faster.

Generation speed ≈ memory bandwidth ÷ bytes read per token.

Hugging Face model cards (Qwen) · GGUF sizes: unsloth UD-Q4_K_M

What to run · Oct 2026 · 64K context

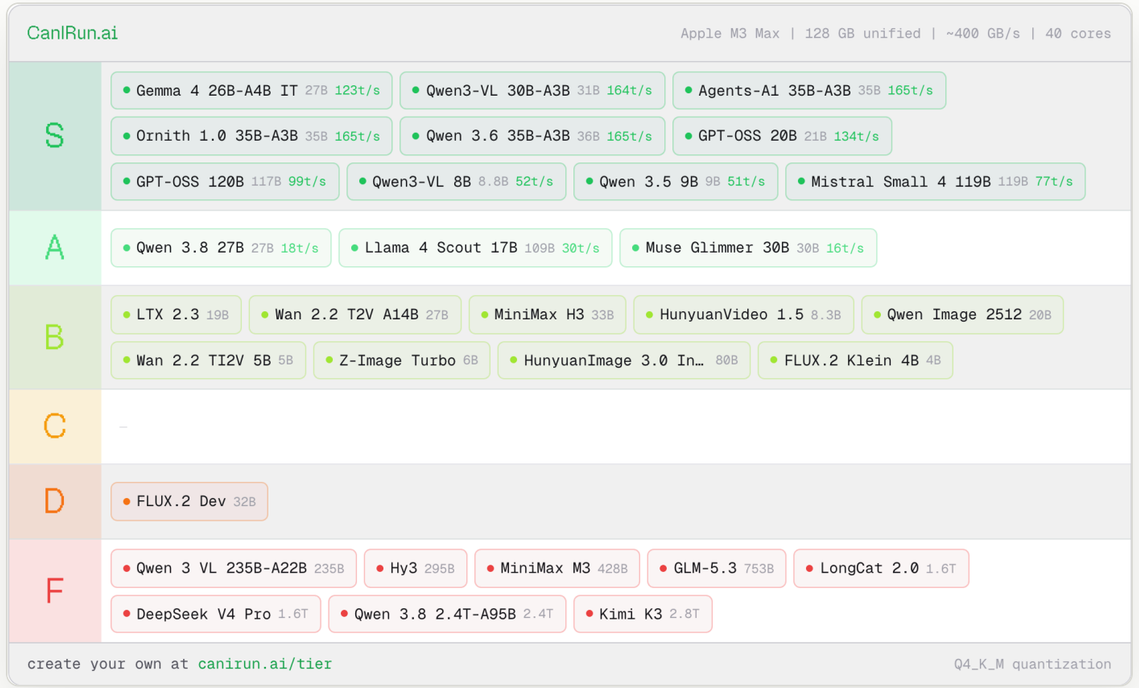
128 GB	Qwen3.8 Flash Next · gpt-oss-120b	90 / 63 GB
64 GB	Qwen3.8-27B at Q8_0	28.6 GB
24–32 GB	Qwen3.6-35B-A3B fast	22.1 GB
	Qwen3.8-27B smarter, slower	16.5 GB
16 GB	Qwen3.5-9B	5.7 GB
8 GB	Qwen3.5-4B	2.7 GB

what I run

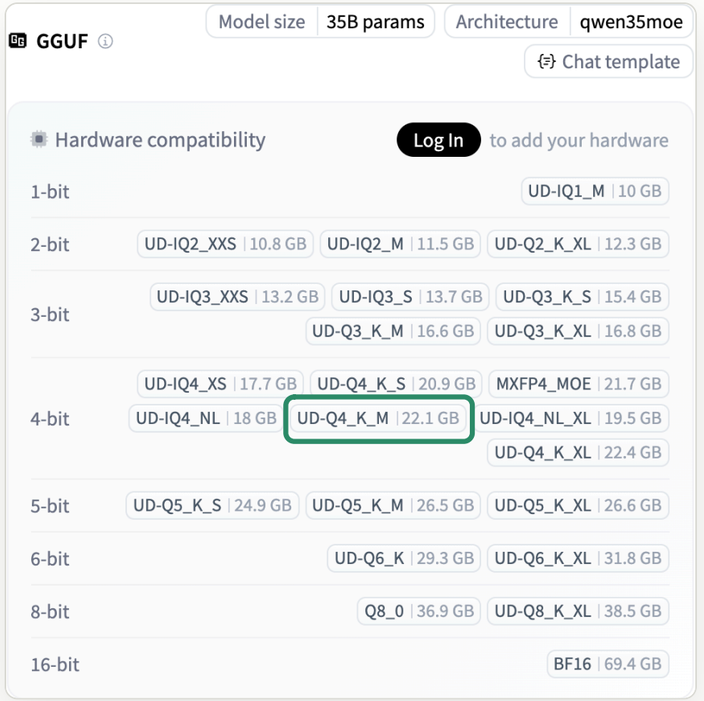
Qwen3.8-27B
UD-Q4_K_M on an M3 Max

tier = memory the GPU can use · GGUF sizes from Hugging Face (unsloth, ggml-org)

Check before you download



canirun.ai/tier · grades models for your machine, S → F



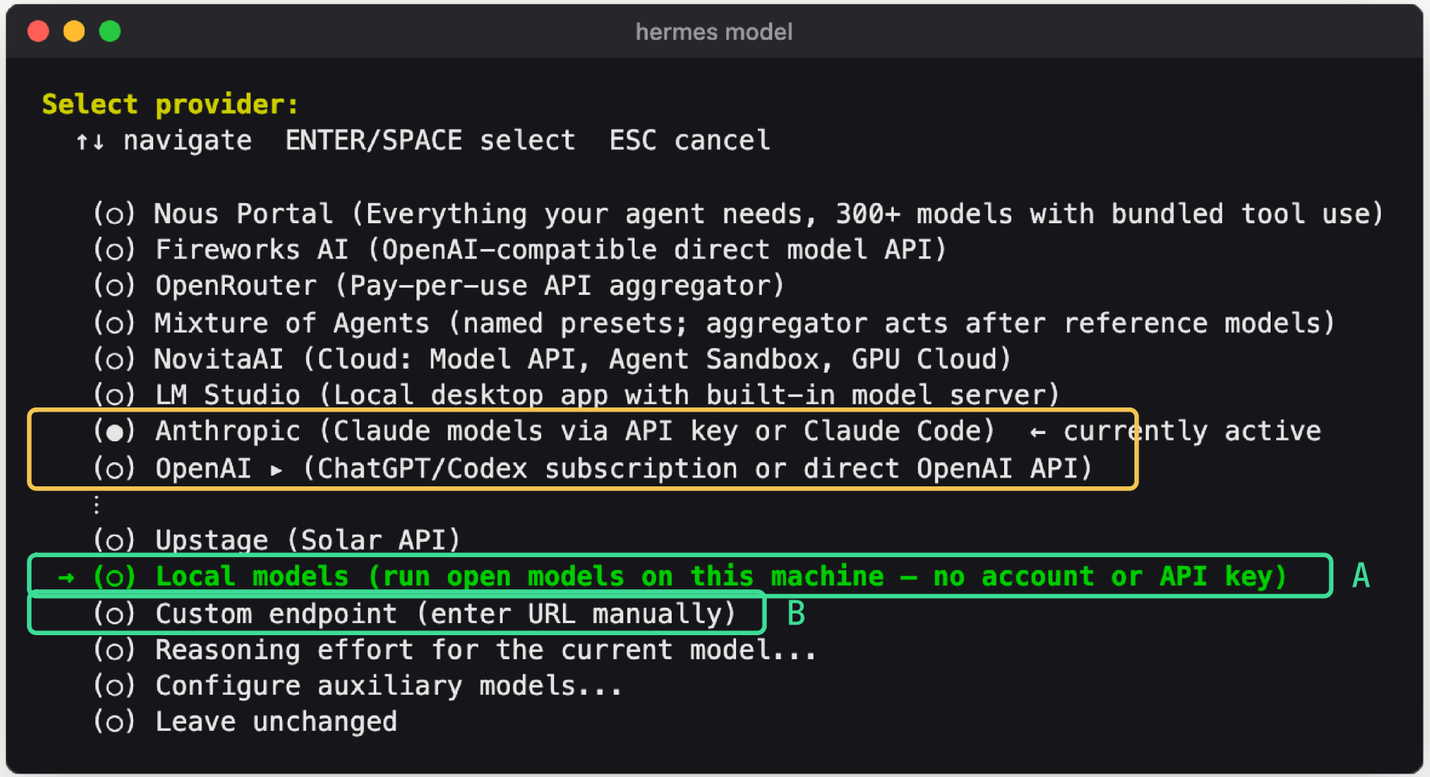
Hugging Face · hardware compatibility

`UD-Q4_K_M`

22.1 GB file ≈ 22 GB of memory

File size on Hugging Face ≈ memory for the weights.

3 · Plug it in: `hermes model`



or attach a subscription

Claude · ChatGPT/Codex · Copilot · Nous Portal

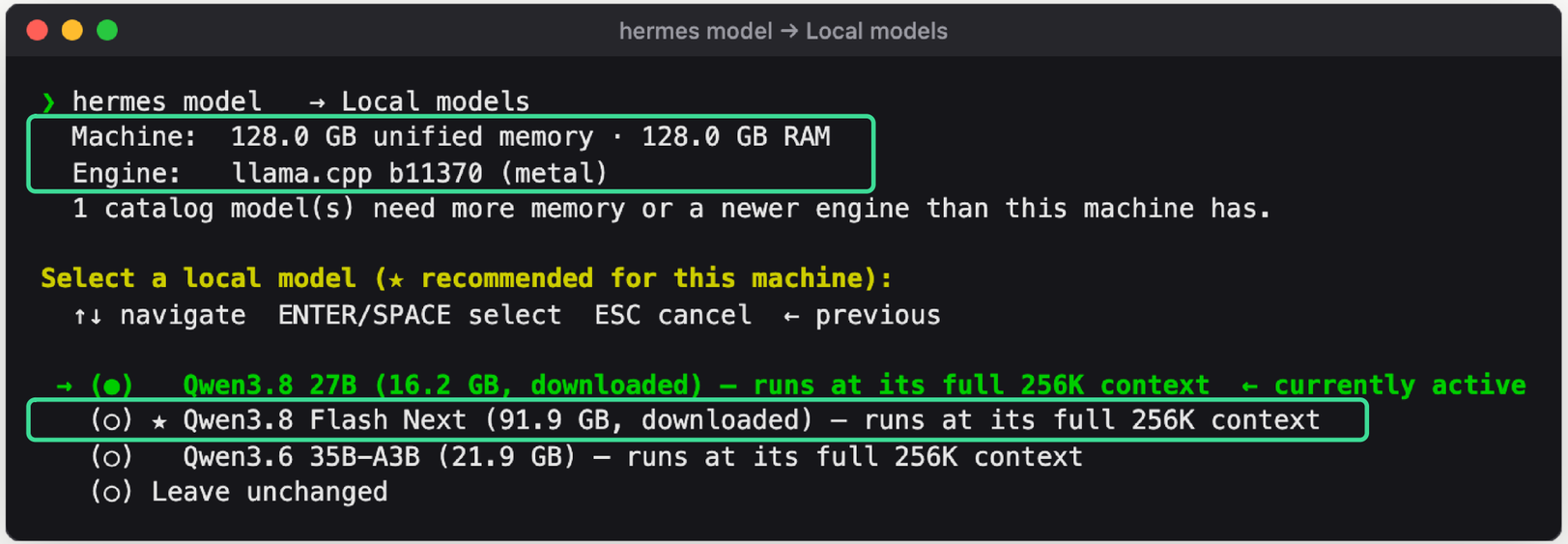
A · local models

Hermes picks, downloads and runs a model for you.

B · custom endpoint

You run the server. Hermes connects to it.

A · Local models: Hermes does it for you

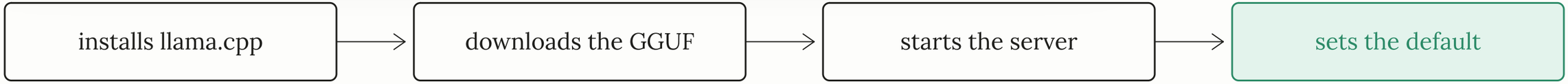


`it measures your machine`

memory, GPU, engine build

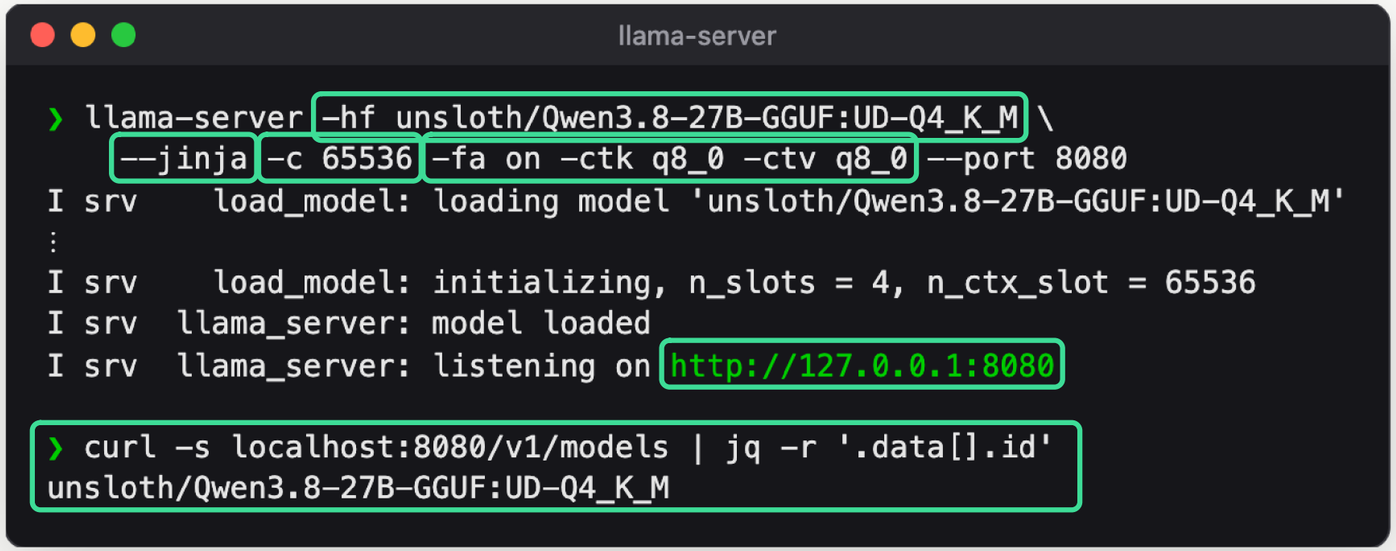
`★ recommended`

best model that fits fully in memory at ≥ 20 tok/s



Mac under 32 GB or GPU under 24 GB → no ★. Use path B with a smaller model.

B · Serve it yourself



Install: `brew install llama.cpp` · Windows / Linux: prebuilt binaries on GitHub

`-hf repo:quant`

downloads the GGUF from Hugging Face

`--jinja`

tool calls

`-c 65536`

Hermes needs ≥ 64K context

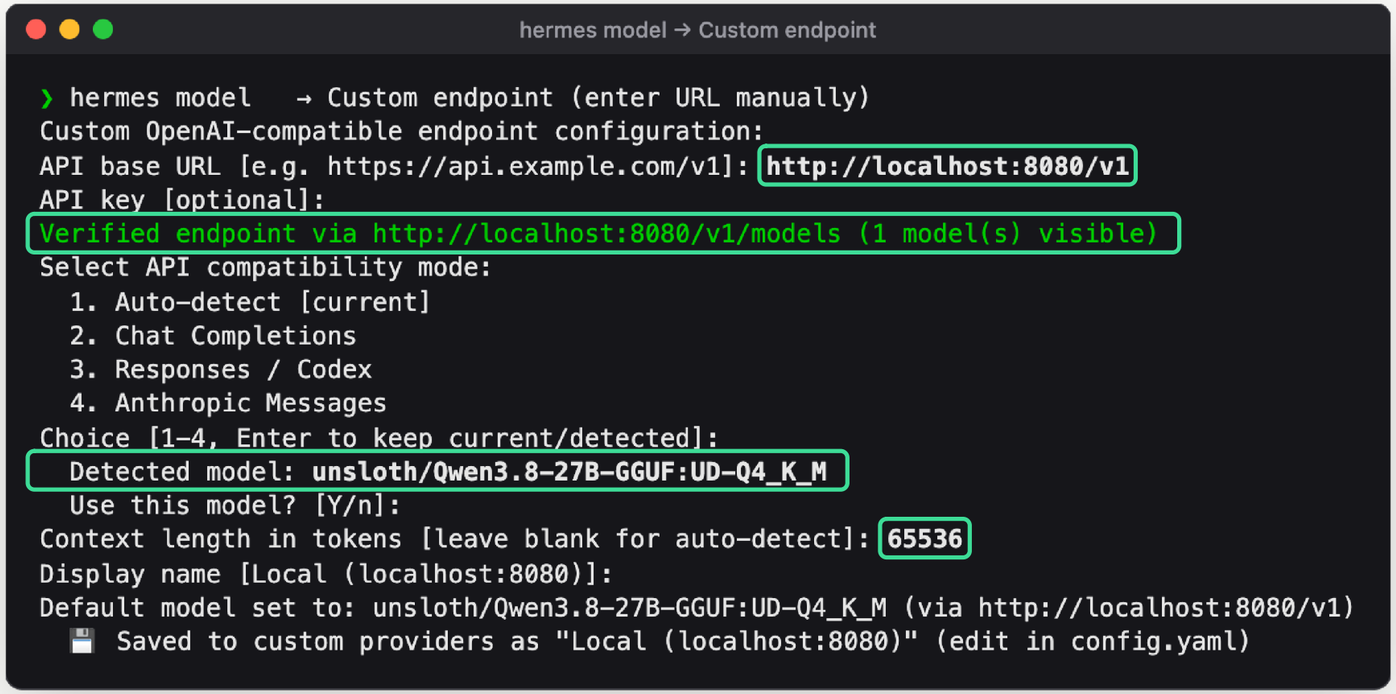
`-ctk/-ctv q8_0 · -fa on`

KV cache at half the memory

`curl …/v1/models`

if this answers, Hermes can connect

B · Point Hermes at it




```yaml
# ~/.hermes/config.yaml (excerpt)
model:
  provider: custom
  default: unsloth/Qwen3.8-27B-GGUF:UD-Q4_K_M
  base_url: http://localhost:8080/v1
custom_providers:
  - name: Local (localhost:8080)
    models:
      unsloth/Qwen3.8-27B-GGUF:UD-Q4_K_M:
        context_length: 65536
```

Edit the file directly any time. `hermes model` just writes it for you.

① base URL
ends in `/v1`

② Hermes checks it
and finds the model

③ context
65536 or more

Same recipe, other servers

Ollama

```
OLLAMA_CONTEXT_LENGTH=65536 ollama serve
```

→ `http://localhost:11434/v1` · its default context is 4K on small GPUs

LM Studio

```
hermes model  →  LM Studio
```

→ `http://127.0.0.1:1234/v1` · set the context to 65536 when you load the model

vLLM

```
vllm serve <model> --max-model-len 65536 \
  --enable-auto-tool-choice --tool-call-parser hermes
```

→ `http://localhost:8000/v1` · NVIDIA servers

Hermes docs: integrations/providers · local-ollama-setup

Prefer a subscription?

Same command, `hermes model` → Nous Portal · ChatGPT / Codex · Claude · GitHub Copilot

switch mid-session

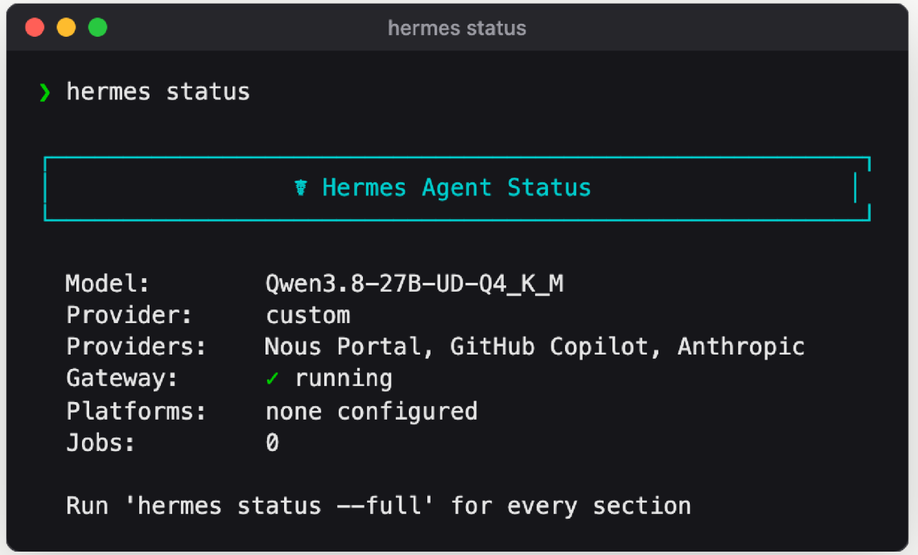
```
/model <name>
```

local first, cloud as backup

```
hermes fallback add
```

check what's active

```
hermes status
```



Four things that break it

`1 · context too small`

Hermes refuses to start below 64K.

→ `-c 65536` · Ollama: `OLLAMA_CONTEXT_LENGTH`

`2 · no tool calls`

The model writes text instead of calling tools.

→ `--jinja` · vLLM: `--tool-call-parser`

`3 · endpoint 404`

The base URL is missing `/v1`.

→ `http://localhost:8080/v1`

`4 · wrong model`

Picking ~~a “Hermes 4” model~~ for Hermes.

Nous Hermes 3/4 are chat models, not agentic. Use Qwen, Gemma, GLM or gpt-oss.

error messages and warnings from the Hermes v0.22.1 source

Run `hermes`



`tools`

files, terminal, web, browser, memory, cron, delegation…

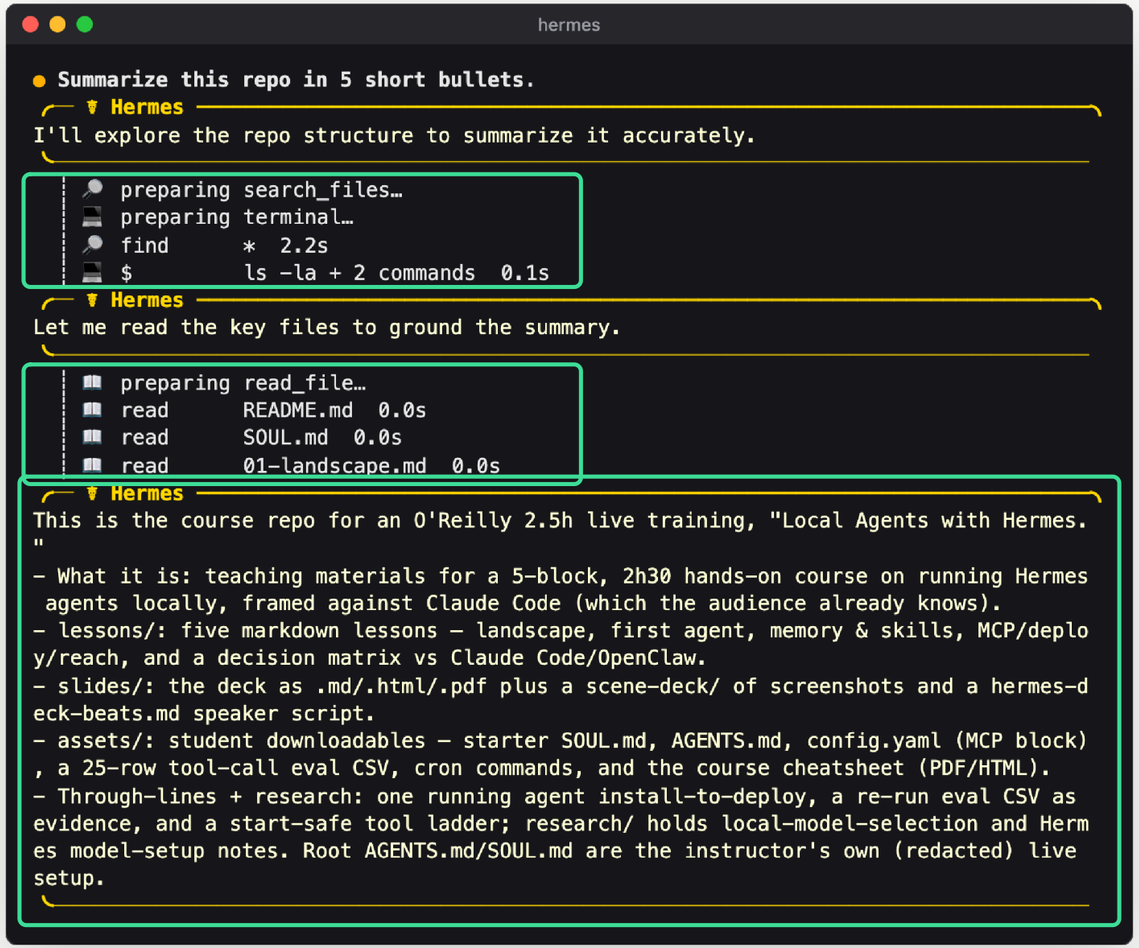
`skills`

58 built in: GitHub, Obsidian, Google Workspace, PDFs…

`model`

the local Qwen you just served

Give it a task



`prompt`

"Summarize this repo in 5 short bullets."

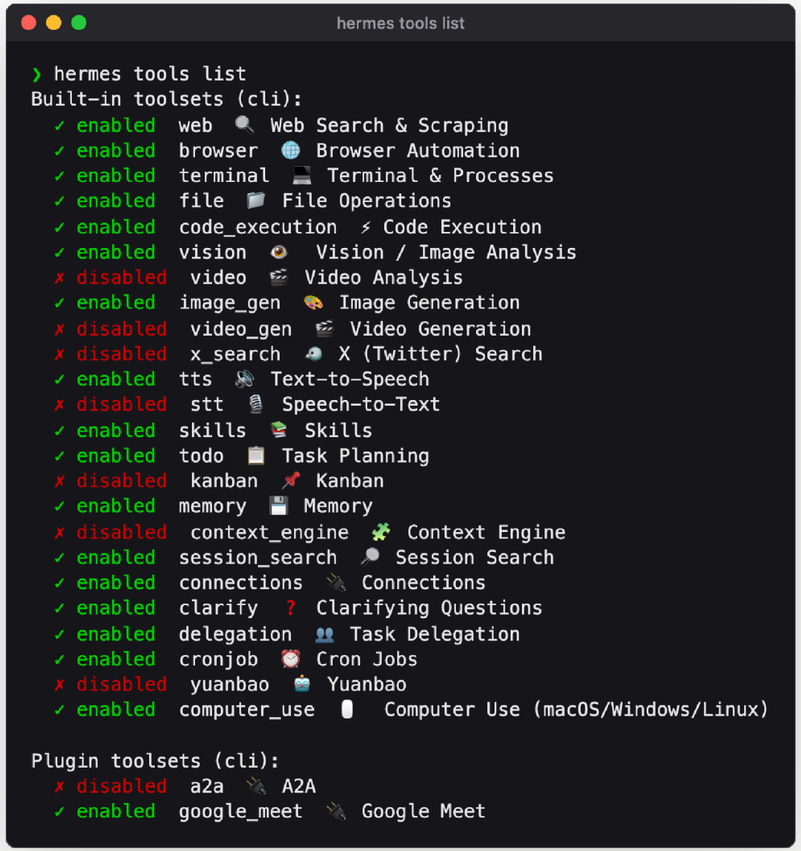
`tool calls`

find · ls · read README.md — it looks before it answers

`answer`

Grounded in the files it read. Local model, $0.

Tools: turn on what you need



Toggle with `hermes tools`, or `/tools` in chat. Widen in this order:

`4 · irreversible`
delete, send, spend · worst case: you explain it to someone

`3 · reversible`
edits inside git · worst case: `git checkout .`

`2 · drafts`
new files, proposed commands · worst case: a junk file

`1 · read-only`
read files, search · worst case: wasted tokens

Integrations are plugins

hermes plugins

```
❯ hermes plugins enable google_meet
✓ Plugin google_meet enabled. Takes effect on next session.
Gateway reloaded plugins — active in the running gateway now: hooks; deferred: tools (next
session).
```

google_meet gives the agent new tools

`meet_join` `meet_say` `meet_leave` `live captions`

more plugins

`hermes plugins browse`

any MCP server

`hermes mcp`

Skills: procedures it can reuse

hermes skills list

```
❯ hermes skills list
                        Installed Skills
```

Name	Category	Source	Trust	Status
working-with-lucas		local	local	enabled
apple-notes	apple	builtin	builtin	enabled
apple-reminders	apple	builtin	builtin	enabled
findmy	apple	builtin	builtin	enabled
imessage	apple	builtin	builtin	enabled
claude-code	autonomous-ai-agents	builtin	builtin	enabled
codex	autonomous-ai-agents	builtin	builtin	enabled
⋮				
obsidian	note-taking	builtin	builtin	enabled
⋮				

1 hub-installed, 57 builtin, 1 local — 59 enabled, 0 disabled



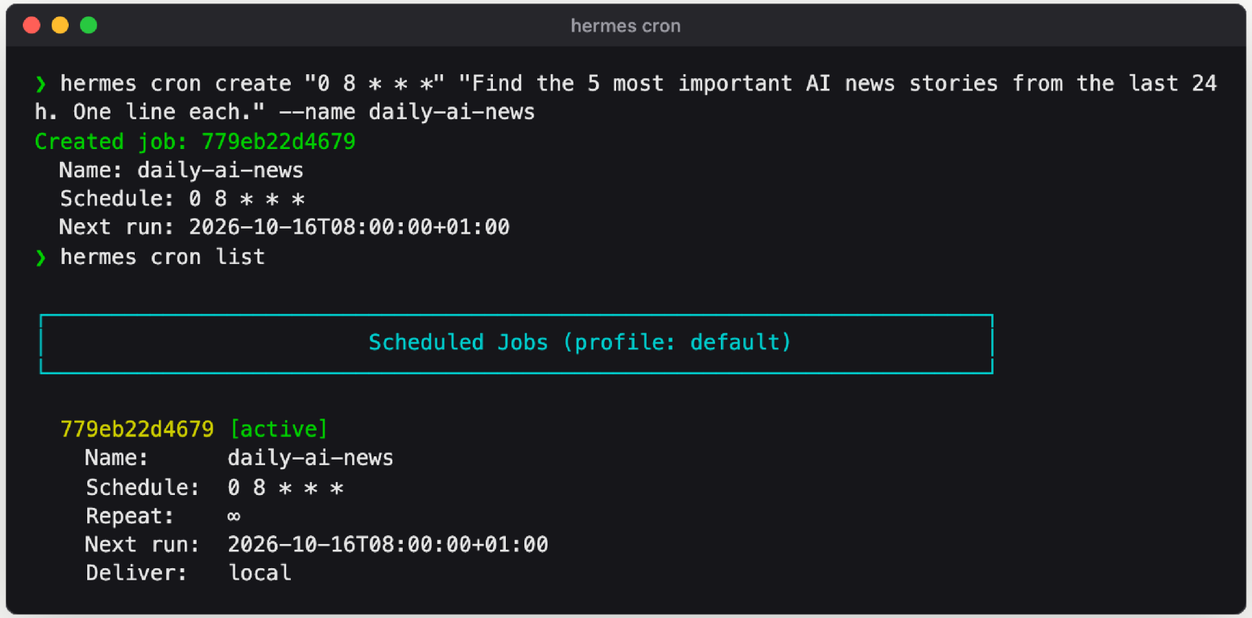
open standard

Same `SKILL.md` format as Claude Code · agentskills.io

it learns

After a hard task, Hermes writes a new skill itself.

Cron: it works while you sleep



`0 8 * * *`

daily AI news digest → Telegram

`0 21 * * *`

Apple Notes → Obsidian

`30 8 * * 1-5`

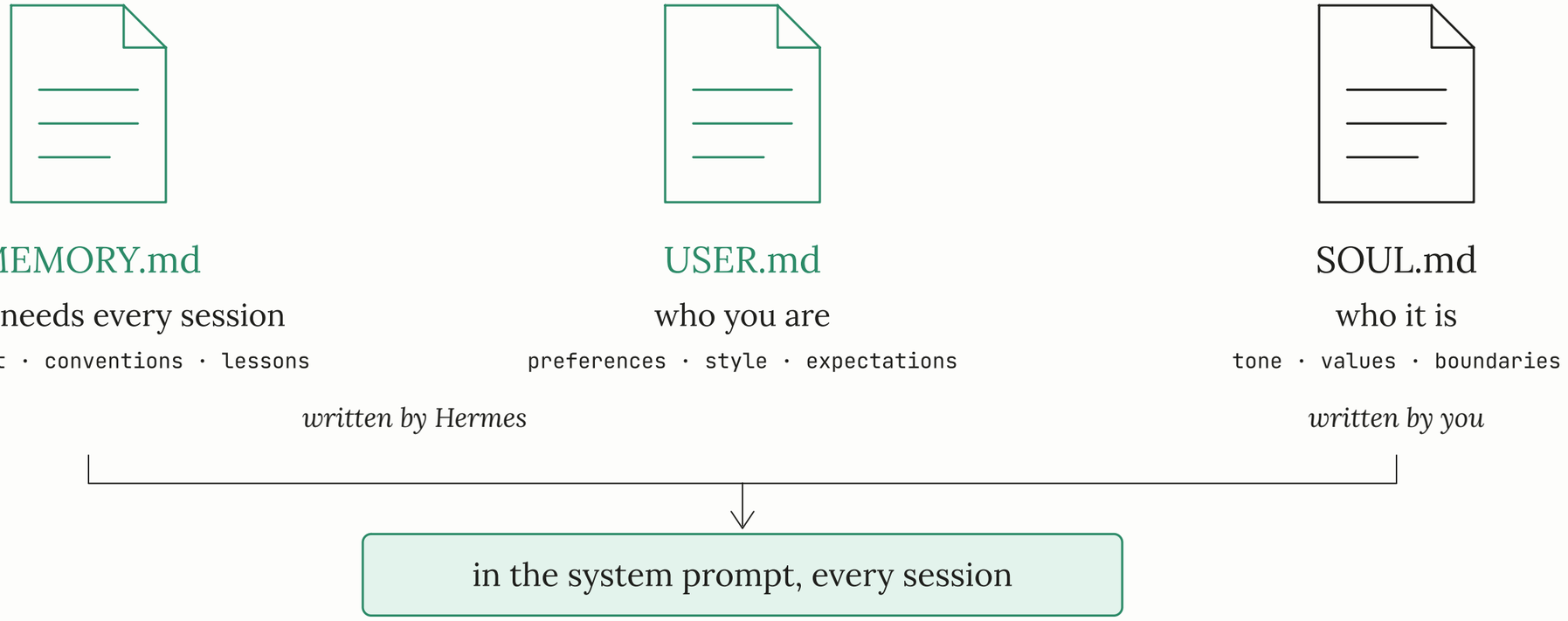
standup prep from yesterday's git log

`0 3 * * *`

nightly health check of your model server

Or just ask in chat: `/cron` “every weekday at 8, summarize my calendar”

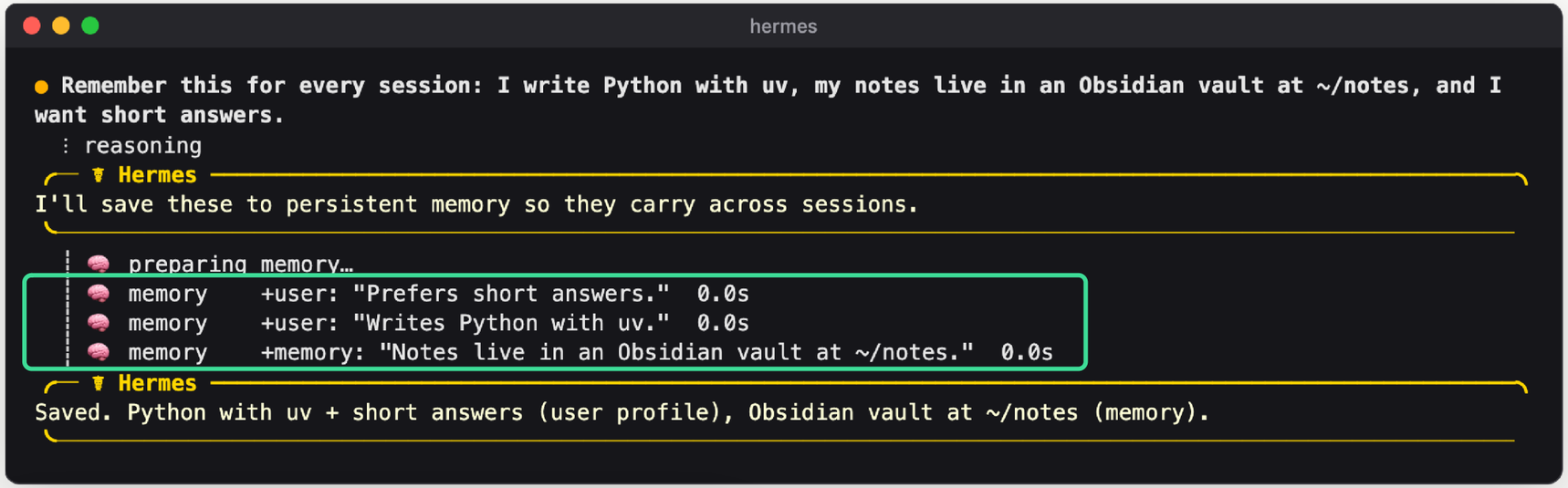
Memory is three small files



Small on purpose (~2,200 + ~1,375 characters). When full, it consolidates.

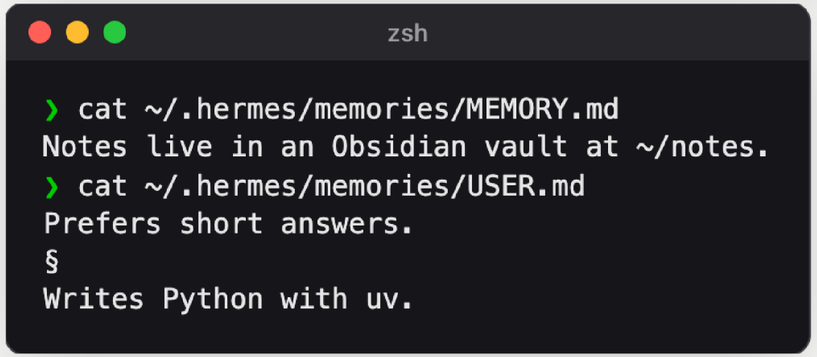
hermes-agent.nousresearch.com/docs/user-guide/features/memory

Memory in action



it decides

preferences → USER.md
environment → MEMORY.md



next session

both files load into the prompt. Nothing to re-explain.

Give it a personality: `SOUL.md`

```
# ~/.hermes/SOUL.md
## Who you are
A direct, technically careful collaborator.

## How you talk
- Lead with the answer.
- Never open with "You're absolutely right".
- Disagree in one sentence, with the reason.

## Boundaries
- Show destructive commands before running them.
- Say "I don't know" instead of guessing.
```

Personality is a file.

Swap the model, keep the soul.

Starter: `assets/SOUL.md` in the course repo

Obsidian as the second brain

`1 · create a vault`

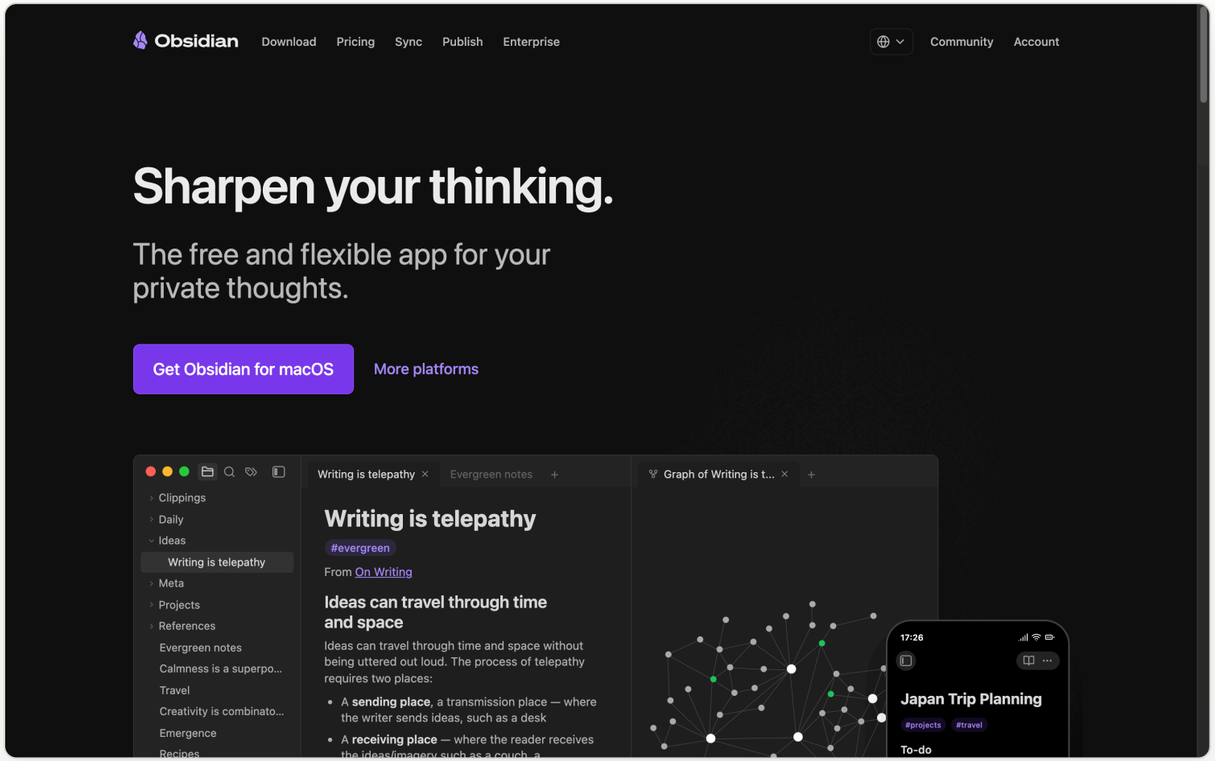
a folder of plain markdown files

`2 · tell Hermes where it is`

“my notes live in ~/notes” → saved to memory

`3 · use the obsidian skill`

built in: search, read, write and link notes



What to save there

`capture`

meeting notes

research reports & summaries

web clippings

daily notes

`build`

project files

PRDs for apps & prototypes

workflows & SOPs

decisions log

`think`

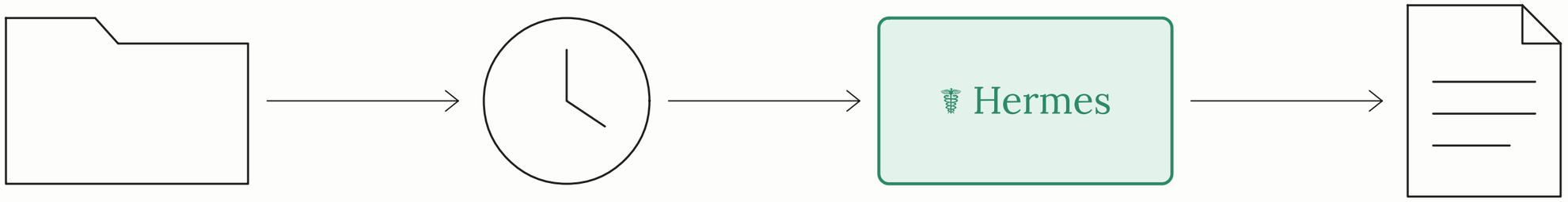
concept & synthesis pages

atomic / evergreen notes

session memory

Rule of thumb: if you'll want it in three months, it goes in the vault.

Meeting notes on autopilot



Meet Recordings

`Google Drive folder`

cron · every evening

summary + action items

a note in the vault

`linked to “Project - …”`

```
hermes cron create "0 19 * * *" "Turn new recordings in ~/Meet Recordings into meeting notes in ~/notes" --skill obsidian
```

Talk to it from your phone



`hermes whatsapp`

pick bot or self-chat, scan the QR (Linked Devices)

`WHATSAPP_ALLOWED_USERS=<your number>`

everyone else gets ignored

`hermes gateway install`

runs in the background, survives reboots

Unofficial bridge: use a dedicated number.

WhatsApp Setup

Hermes connects to WhatsApp through a built-in bridge based on **Baileys**. This works by emulating a WhatsApp Web session — **not** through the official WhatsApp Business API. No Meta developer account or Business verification is required.

> Run `hermes gateway setup` and pick **WhatsApp** for a guided walk-through.

TWO WHATSAPP INTEGRATIONS

This page is for the **Baileys bridge** — quick to set up, personal accounts, no public URL needed, ban risk.

If you're running a real business bot and want stability, see the **WhatsApp Business Cloud API guide** instead. It's the official Meta-supported path: no account ban risk, but requires a Meta Business account and a public webhook URL.

The two adapters can also run in parallel against different phone numbers if you have a reason to.

UNOFFICIAL API — BAN RISK

WhatsApp does **not** officially support third-party bots outside the Business API. Using a third-party bridge carries a small risk of account restrictions. To minimize risk:

- **Use a dedicated phone number** for the bot (not your personal number)
- **Don't send bulk/spam messages** — keep usage conversational
- **Don't automate outbound messaging** to people who haven't messaged first

hermes-agent.nousresearch.com/docs/user-guide/messaging/whatsapp

Workflow tips



1 · front-load context

Paste the error, the path, the screenshot (Ctrl+V). Then ask.



2 · resume, don't restart

`hermes -c` picks up the last session.



3 · twice → skill · three times → cron

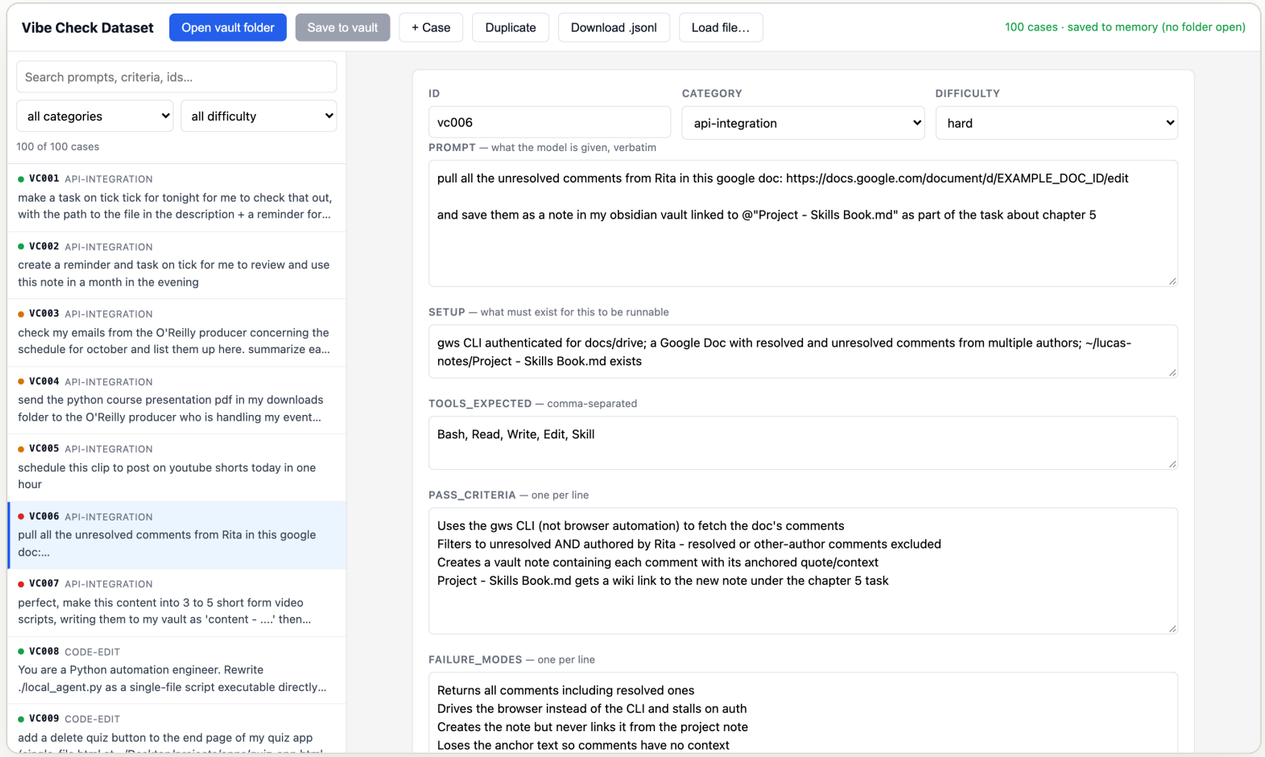
Repeated work becomes a procedure, then a schedule.



4 · start read-only

Give it more tools as it earns your trust.

Is the local model good enough? Vibe-check it.



`100 cases`

real tasks from my own agent history

`each case`

prompt · tools expected · pass criteria · failure modes

`obsidian-vault`	
`shell-automation`	
`media-pipeline`	
`code-edit`	
`multi-step`	
`search-retrieval`	

+7 more categories · score each one, not one number

Swap the model. Keep the harness. Re-run.

When to use what

Hermes

your standing personal agent

memory · cron · messaging · any model

Claude Code

a deep coding session in one repo

the most mature coding tooling

OpenClaw

the lightest CLI coding agent

minimal setup

Pick Hermes when the accumulated context is the point.

Your model is swappable.

Your context compounds.

course repo

github.com/EnkrateiaLucca/hermes-course

docs

hermes-agent.nousresearch.com/docs